Exercise 7: Plant tissue culture – Callus with plantlets

Aim: To study and identify the Callus with plantlets.

Principle: Growing the plant cells, tissues and organs in an artificial, synthetic medium under controlled conditions is called plant tissue culture. The technique of cloning plant is easier than animals because plant cells are simple in structure and most plant cells shows totipotency (i.e) ability to regenerate from cells.

Requirements: Model / Photograph / Picture of callus with plantlets.

Diagnostic Features

- The callus is an unorganized mass of undifferentiated tissue.
- The mechanism of callus formation is that auxin induce cell elongation and cytokinin induces cell division as a result of which masses of cells are formed.
- Roots and shoots are differentiated from the callus.

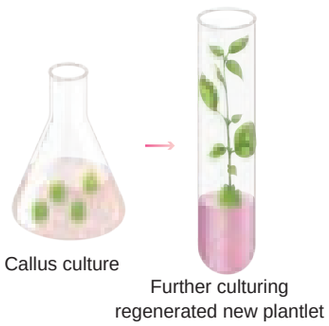


Figure 7: Callus with plantlets

Exercise 8: Types of ecological pyramid

Aim: To study and identify the different types of ecological pyramids

Principle: The relationship between different trophic levels in an ecosystem when shown diagrammatically appear as 'ecological pyramids'. In these ecological pyramids, the successive tiers represent successive trophic levels towards the apex. The base of the pyramid is of producers, the next one above it is of herbivores and the top tiers are of carnivores. The top most or apex represents the tertiary or top level consumers.

Requirements: Models / Photographs / Pictures of different types of ecological pyramid.

8 A. Pyramid of numbers

Diagnostic Features

- The number of organism that are present in successive trophic levels of an ecosystem is shown in the pyramid of numbers of a grassland ecosystem.
- There is a gradual decrease in the number of organisms in each trophic level from producers to primary consumers, then to secondary consumer, and finally to tertiary consumers.

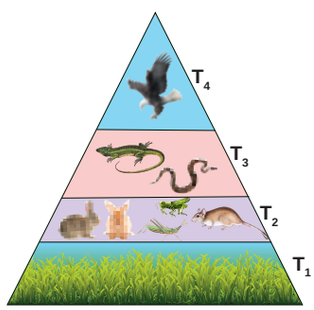


Figure 8 a: Pyramid of numbers in grassland ecosystem

- Therefore, pyramid of number in grassland ecosystem is always upright.

T_1 - Producers | T_2 - Herbivores | T_3 - Secondary consumers | T_4 - Tertiary consumers

TN_GOVT_XII_BIOLOGY_BOTANY_PRACTICAL_MANUAL_PAGES_239-258.indd 230 3/14/2020 12:13:03 PM